



BKBPS TEACHERS

Capstone Project Presentation



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Presentation Tool: Canva

Problem Statement



Many people use single-use plastic daily because it is convenient and cheap. How might we design affordable and sustainable alternatives or systems that encourage people to reduce their dependence on single-use plastic?



1. Many Interconnected Causes



2. Multiple Stakeholders



3. No Single "Correct" Solution



WHY IS IT A
COMPLEX
PROBLEM?



4. Behaviour Change is required

5. Trade-offs Between
Cost, Convenience, and
Sustainability



6. Solutions Must Work in
Real Contexts





By the end of the project, students will:

PRESENT

a tested solution to reduce plastic waste in their school or community.

DEMONSTRATE

the Design Thinking process from empathy to testing.

REFLECT

on how individual actions can contribute to environmental sustainability.



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Goal



Students will investigate the problem of plastic waste in their community and design practical, user-centered solutions that reduce the use and impact of single-use plastics.



Table 1 – Project Planning (Design Thinking Model)

Phase	Stage	Activities
Scope	Problem Exploration	• Brainstorm sources of single-use plastic in daily life (bags, bottles, packaging, straws). • Conduct waste audit in school and nearby market to identify major plastic waste sources. • Study statistics on plastic waste generation in India. • Create a visual map of plastic usage areas (home, school, shops, restaurants). • Review case studies from the cleanest cities such as Singapore, Copenhagen and Tokyo.
Empathize	Stakeholder & User Research	• Conduct surveys among households, students and shopkeepers to understand plastic usage behaviour. • Interview waste collectors to understand challenges in plastic recycling. • Observe consumer habits in local markets and supermarkets. • Create stakeholder mapping (consumers, businesses, government, recyclers). • Develop an empathy map to understand motivations, needs and problems of users.
Define	Problem Analysis	• Analyze survey data to identify the main causes of plastic dependency. • Create cause-effect (fishbone) diagram to identify root problems. • Identify gaps in current waste management systems. • Compare waste practices in India with those of clean cities. • Write a clear problem statement describing the challenge.





Project planning (DT Model)

PHASE	STAGE	ACTIVITIES
Ideate	Solution Generation	• Brainstorm innovative ideas to reduce plastic use. • Study global solutions like bottle deposit systems, refill stations and waste-to-energy plants. • Generate alternative materials (cloth bags, biodegradable packaging, reusable containers). • Conduct group discussions to evaluate feasibility of solutions. • Prepare concept sketches and system diagrams for selected ideas.
Prototype	Model Development	• Create a prototype such as cloth bag bank, bottle return station, plastic collection box or refill station model. • Design posters, awareness materials and process flow diagrams. • Develop a working model demonstrating the plastic reduction system. • Test usability of the prototype with students and teachers.
Validation	Testing & Feedback	• Implement the prototype in a small area (school or classroom). • Collect feedback from users regarding convenience and effectiveness. • Measure reduction in plastic waste during the trial period. • Identify limitations and areas for improvement.





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Project planning (DT Model)

PHASE

STAGE

ACTIVITIES

Implementation

Real
Application

- Implement the improved system in school or nearby community.
- Install segregation bins and plastic collection stations.
- Introduce reusable bag systems and plastic return incentives.
- Conduct awareness campaigns and workshops.

Report

Documentation
& Reflection

- Document research findings, survey results and global case studies.
- Analyze the effectiveness of implemented solutions.
- Present results through charts, graphs and photographs.
- Provide recommendations for scaling the solution to larger communities.



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Requirements & Specifications



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Solution Model



PHASE WISE PLANNING



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